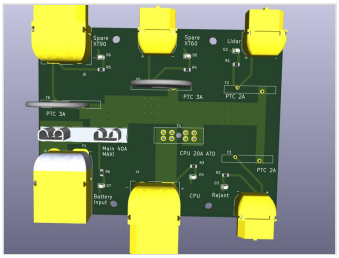
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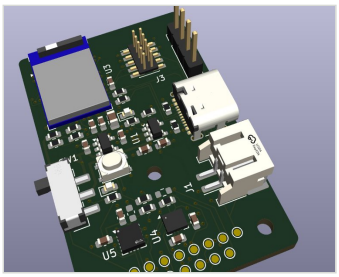
Projects



Spot Power Distribution Board

Team PRONTO, GRASP Lab at Penn. DARPA Triage Challenge, summer 2026.

Designed the board that feeds compute, lidar and the mesh radio on PRONTO's Spot robots. A **40 A main fuse** splits into five individually fused branches, and the **20 A compute rail** runs as a shaped pour on 2 oz copper rather than a trace. Schematic, layout, indicators and BOM in KiCad, through design review and out to fab with my name on the silkscreen.



Golf Swing Sensor Board

My own project. Designed 2026, assembled at JLCPCB.

A **4-layer, 30 by 45 mm** board that measures a golf swing and sends it over Bluetooth. An nRF52840 in a pre-certified module does radio and compute, a 6-axis ICM-45686 covers the swing, and a **200 g ADXL372** catches the impact spike the first sensor would saturate on. I set those ranges from a ball flight model I wrote rather than guessing, then took the board through sourcing, DFM and assembly with a clean DRC.



Autonomous Color Sorter

Four-person team, ENGEK 210 at Boston University. Spring 2025.

Load it with Skittles and walk away. It hits **92% accuracy at 2.62 seconds a Skittle**, against targets of 10% error and 5 seconds. On a four-person team I did the CAD, the circuit schematic and all the wiring, from the TCS34725 color sensor to the servo chute, and laser-cut the electronics housing and the acrylic spacer that fixes the sensor-to-Skittle gap. Bins carry braille as well as printed labels, so using it never depends on color vision.



Status Lighting and Link Failover, Jackal Fleet

Team PRONTO, GRASP Lab at Penn. Summer 2026.

Two pieces of operator plumbing on robots other people were running. Status lights that say which robot is which and whether it is safe to walk up to, and a Rajant-to-WiFi failover for RTK corrections that **ships disabled by default**, so a live field system could take the merge with no risk.

Also building

Self-driving track robot
An ESP32 and lidar robot learning to drive a track by imitating me, then collecting its own correction data wherever the policy is weakest. Dual-core firmware, 20 Hz control loop, Python ground station. In hardware bring-up.

Random positioning machine
A two-axis bench instrument that keeps reorienting a sample so its gravity vector averages toward zero, approximating microgravity for cell work. Assembled, both axes turning in early tests, paused on a power fault.

Hardware	KiCad, schematic capture, 2 and 4-layer layout, power, BOM, DFM, JLCPCB	*Embedded*	nRF52840, ESP32, Arduino, I2C, UART, BLE, sensors, motor drive
Software	ROS 2, Python, Linux, Docker, Git	*Shop*	SolidWorks, Fusion 360, dimensioned drawings, 3D printing, laser cutting

Full project sheets with drawings, schematics and results available on request.

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